**TSM 098 | INDEPENDENT PRACTICE**

# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

## Core: combine fractions on each side

**1.** Solve and check in the original equation.

$$(\frac{3x+4}{6}) + (\frac{x-3}{2}) = (\frac{2x-1}{4})$$

**2.** Solve and check in the original equation.

$$(\frac{-2x+2}{5}) + (\frac{-1}{4}) = (\frac{x+6}{8})$$

**3.** Solve and check in the original equation.

$$(\frac{3x+1}{3}) + (\frac{2}{4}) = (\frac{x+1}{2})$$

**4.** Solve and check in the original equation.

$$(\frac{-3x-5}{2}) + (\frac{-x+4}{8}) = (\frac{-2x-6}{4})$$

**5.** Solve and check in the original equation.

$$(\frac{-2x+2}{6}) + (\frac{x-1}{2}) = (\frac{-2x-6}{3})$$

**6.** Solve and check in the original equation.

$$(\frac{-2x-2}{3}) + (\frac{4}{2}) = (\frac{-x-1}{5})$$

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## Core: numerical denominators on both sides

**7.** Solve and check in the original equation.

$$(\frac{3x-1}{3}) + (\frac{x-1}{2}) = (\frac{x+6}{6})$$

**8.** Solve and check in the original equation.

$$(\frac{x+4}{2}) + (\frac{0}{4}) = (\frac{x+4}{5})$$

**9.** Solve and check in the original equation.

$$(\frac{2x+1}{2}) + (\frac{x-4}{5}) = (\frac{0}{8})$$

**10.** Solve and check in the original equation.

$$(\frac{-2x-3}{8}) + (\frac{3}{6}) = (\frac{-2x+4}{4})$$

**11.** Solve and check in the original equation.

$$(\frac{-2x-4}{2}) + (\frac{x+1}{3}) = (\frac{-6}{6})$$

**12.** Solve and check in the original equation.

$$(\frac{-2x+1}{3}) + (\frac{0}{4}) = (\frac{2}{8})$$

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# Linear equations with numerical fractions

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Name: _____ Date: _____

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## Core: signed numerators and fractions

**13.** Solve and check in the original equation.

$$(\frac{x-4}{2})+(\frac{x-3}{4})=(\frac{6}{6})$$

**14.** Solve and check in the original equation.

$$(\frac{-2x-2}{5})+(\frac{x+1}{8})=(\frac{2}{2})$$

**15.** Solve and check in the original equation.

$$(\frac{x+3}{4})+(\frac{-x-1}{6})=(\frac{-x+5}{3})$$

**16.** Solve and check in the original equation.

$$(\frac{-2x+2}{3})+(\frac{-x}{8})=(\frac{-2x+4}{6})$$

**17.** Solve and check in the original equation.

$$(\frac{-2x-4}{6})+(\frac{-x-4}{4})=(\frac{-2x+3}{2})$$

**18.** Solve and check in the original equation.

$$(\frac{2x+5}{4})+(\frac{x-2}{3})=(\frac{2x+1}{6})$$

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# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

## Extension: mixed fraction equations

**19.** Solve and check in the original equation.

$$\left(\frac{-2x-5}{3}\right)+\left(\frac{4}{2}\right)=\left(\frac{x-4}{4}\right)$$

**20.** Solve and check in the original equation.

$$\left(\frac{-2x-4}{8}\right)+\left(\frac{-x}{6}\right)=\left(\frac{5}{4}\right)$$

**21.** Solve and check in the original equation.

$$\left(\frac{2x+5}{4}\right)+\left(\frac{1}{6}\right)=\left(\frac{4}{3}\right)$$

**22.** Solve and check in the original equation.

$$\left(\frac{x+3}{5}\right)+\left(\frac{0}{3}\right)=\left(\frac{2x+5}{4}\right)$$

**23.** Solve and check in the original equation.

$$\left(\frac{2x+2}{8}\right)+\left(\frac{-x-2}{2}\right)=\left(\frac{-2x+4}{3}\right)$$

**24.** Solve and check in the original equation.

$$\left(\frac{-2x+4}{4}\right)+\left(\frac{-x}{3}\right)=\left(\frac{-2x-2}{2}\right)$$

**TSM 098 | ANSWER KEY**

# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

## Core: combine fractions on each side

**1.** Solve and check in the original equation.

$$\frac{6x-5}{6}=\frac{2x-1}{4}$$

$$4(6x-5)=6(2x-1)$$

$$x=\frac{7}{6}$$

Check by substituting into the original equation.

**2.** Solve and check in the original equation.

$$\frac{-8x+3}{20}=\frac{x+6}{8}$$

$$8(-8x+3)=20(x+6)$$

$$x=-\frac{8}{7}$$

Check by substituting into the original equation.

**3.** Solve and check in the original equation.

$$\frac{6x+5}{6}=\frac{x+1}{2}$$

$$2(6x+5)=6(x+1)$$

$$x=-\frac{2}{3}$$

Check by substituting into the original equation.

**4.** Solve and check in the original equation.

$$\frac{-13x-16}{8}=\frac{-x-3}{2}$$

$$2(-13x-16)=8(-x-3)$$

$$x=-\frac{4}{9}$$

Check by substituting into the original equation.

**5.** Solve and check in the original equation.

$$\frac{x-1}{6}=\frac{-2x-6}{3}$$

$$3(x-1)=6(-2x-6)$$

$$x=-\frac{11}{5}$$

Check by substituting into the original equation.

**6.** Solve and check in the original equation.

$$\frac{-2x+4}{3}=\frac{-x-1}{5}$$

$$5(-2x+4)=3(-x-1)$$

$$x=\frac{23}{7}$$

Check by substituting into the original equation.

**TSM 098 | ANSWER KEY**

# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

## Core: numerical denominators on both sides

**7.** Solve and check in the original equation.

$$\frac{9x-5}{6}=\frac{x+6}{6}$$

$$6(9x-5)=6(x+6)$$

$$x=\frac{11}{8}$$

Check by substituting into the original equation.

**8.** Solve and check in the original equation.

$$\frac{x+4}{2}=\frac{x+4}{5}$$

$$5(x+4)=2(x+4)$$

$$x=-4$$

Check by substituting into the original equation.

**9.** Solve and check in the original equation.

$$\frac{12x-3}{10}=0$$

$$1(12x-3)=10(0)$$

$$x=\frac{1}{4}$$

Check by substituting into the original equation.

**10.** Solve and check in the original equation.

$$\frac{-2x+1}{8}=\frac{-x+2}{2}$$

$$2(-2x+1)=8(-x+2)$$

$$x=\frac{7}{2}$$

Check by substituting into the original equation.

**11.** Solve and check in the original equation.

$$\frac{-2x-5}{3}=-1$$

$$1(-2x-5)=3(-1)$$

$$x=-1$$

Check by substituting into the original equation.

**12.** Solve and check in the original equation.

$$\frac{-2x+1}{3}=\frac{1}{4}$$

$$4(-2x+1)=3(1)$$

$$x=\frac{1}{8}$$

Check by substituting into the original equation.

**TSM 098 | ANSWER KEY**

# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

## Core: signed numerators and fractions

**13.** Solve and check in the original equation.

$$\frac{3x-11}{4}=1$$

$$1(3x-11)=4(1)$$

$$x=5$$

Check by substituting into the original equation.

**14.** Solve and check in the original equation.

$$\frac{-11x-11}{40}=1$$

$$1(-11x-11)=40(1)$$

$$x=-\frac{51}{11}$$

Check by substituting into the original equation.

**15.** Solve and check in the original equation.

$$\frac{x+7}{12}=\frac{-x+5}{3}$$

$$3(x+7)=12(-x+5)$$

$$x=\frac{13}{5}$$

Check by substituting into the original equation.

**16.** Solve and check in the original equation.

$$\frac{-19x+16}{24}=\frac{-x+2}{3}$$

$$3(-19x+16)=24(-x+2)$$

$$x=0$$

Check by substituting into the original equation.

**17.** Solve and check in the original equation.

$$\frac{-7x-20}{12}=\frac{-2x+3}{2}$$

$$2(-7x-20)=12(-2x+3)$$

$$x=\frac{38}{5}$$

Check by substituting into the original equation.

**18.** Solve and check in the original equation.

$$\frac{10x+7}{12}=\frac{2x+1}{6}$$

$$6(10x+7)=12(2x+1)$$

$$x=-\frac{5}{6}$$

Check by substituting into the original equation.

**TSM 098 | ANSWER KEY**

# Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

## Extension: mixed fraction equations

**19.** Solve and check in the original equation.

$$\frac{-2x+1}{3} = \frac{x-4}{4}$$

$$4(-2x+1) = 3(x-4)$$

$$x = \frac{16}{11}$$

Check by substituting into the original equation.

**20.** Solve and check in the original equation.

$$\frac{-5x-6}{12} = \frac{5}{4}$$

$$4(-5x-6) = 12(5)$$

$$x = -\frac{21}{5}$$

Check by substituting into the original equation.

**21.** Solve and check in the original equation.

$$\frac{6x+17}{12} = \frac{4}{3}$$

$$3(6x+17) = 12(4)$$

$$x = -\frac{1}{6}$$

Check by substituting into the original equation.

**22.** Solve and check in the original equation.

$$\frac{x+3}{5} = \frac{2x+5}{4}$$

$$4(x+3) = 5(2x+5)$$

$$x = -\frac{13}{6}$$

Check by substituting into the original equation.

**23.** Solve and check in the original equation.

$$\frac{-x-3}{4} = \frac{-2x+4}{3}$$

$$3(-x-3) = 4(-2x+4)$$

$$x = 5$$

Check by substituting into the original equation.

**24.** Solve and check in the original equation.

$$\frac{-5x+6}{6} = -x-1$$

$$1(-5x+6) = 6(-x-1)$$

$$x = -12$$

Check by substituting into the original equation.